

I am a Senior Data Engineer w/ strong experience in designing, building and maintaining end-to-end data pipelines using Python, SQL and Spark. Over the 10 years, I've worked on creating efficient data models, improving data quality and automating ETL workflows w/ Airflow and dbt to support analytics and reporting needs. I've hands-on experience w/ cloud platforms like AWS and BigQuery, focusing on performance optimization, cost efficiency and scalability. I enjoy solving complex data challenges, collaborating with cross-functional teams and finding practical ways to make data more reliable and meaningful for business growth.

Education

Bachelor's Degree in Computer Science

Stanford University - Palo Alto, CA | Jun 2015

Skills

- Languages: Python, JavaScript, TypeScript, Java, SQL
- Markup & Data Formats: HTML, CSS, JSON, YAML, GraphQL, Protobuf
- Frameworks: Django, Flask, Spring, React, Angular, Android,
- DevOps & Infrastructure: Docker, Kubernetes, Helm, Jenkins, Ansible, Terraform, Git, Kubernetes Operators, ChatGPT Integrations
- Data & Streaming Technologies: Apache Kafka, MQTT Brokers, Elasticsearch, Prometheus, Apache Airflow
- Databases: MySQL, PostgreSQL, Redis, InfluxDB, MongoDB, Cassandra, DynamoDB
- Data Warehouses: BigQuery, Databricks, Amazon Redshift, ClickHouse, DuckDB, Apache Druid
- Networking & APIs: gRPC, GraphQL, HTTP, WebSockets
- Data Science & Machine Learning: Apache Spark, Pandas, NumPy, SciPy, TensorFlow, Keras, scikit-learn, OpenCV, NLTK
- Analytics & Visualization: Tableau, Power BI, AWS QuickSight, Microsoft Excel, Redash, Matplotlib, DeFi & On-Chain Analytics
- AI & MLOps: Large Language Models (LLMs), LangChain, LlamaIndex, MLflow, Kubeflow, PyTorch, FastAPI
- Cloud Platforms: AWS, Google Cloud Platform (GCP), Microsoft Azure

Professional Experience

IBM – Armonk, NY

Senior Data Engineer (Jun 2022 - Current)

Project: Real-Time Analytics Platform for Cloud Operations

- Designed and led the development of a distributed real-time analytics platform that ingested and visualized cloud infrastructure telemetry for a Fortune 500 client.
- Architected a high-throughput data streaming ecosystem using Apache Kafka, processing billions of telemetry events daily.
- Developed distributed Python microservices for metric collection, enrichment, and aggregation from hybrid cloud environments.
- Engineered schema design for time-series and operational metrics leveraging Prometheus and ClickHouse for high-performance storage and analytics.
- Built adaptive streaming pipelines using Kafka Streams (Java) and Faust (Python) for heterogeneous telemetry normalization.
- Implemented FastAPI and gRPC-based query services enabling low-latency analytics across global operations.
- Deployed a fully containerized Kubernetes environment with Helm and Istio for service mesh routing, observability, and security.
- Integrated CI/CD pipelines with Jenkins and Terraform, enforcing environment parity and automated multi-region deployments.
- Defined multi-tier data retention strategies balancing cost, performance, and compliance requirements.
- Introduced data contract governance via Confluent Schema Registry for schema evolution tracking.
- Enhanced Kafka performance through optimized partitioning, replication, and consumer lag monitoring.
- Created Grafana-based dashboards providing real-time visibility into system latency, utilization, and SLA compliance.
- Integrated MLflow-based anomaly detection models and Prometheus alert rules for proactive incident management.
- Collaborated with SREs to define SLIs, SLOs, and automated alerting thresholds.
- Implemented OAuth2-secured APIs and RBAC-based data access policies for multi-tenant data isolation.
- Developed a Python SDK and CLI utilities to standardize metric publishing and schema registration across teams.
- Enabled full lineage tracing by embedding correlation IDs across services and Kafka topics.

- Partnered with data science teams to operationalize predictive anomaly scoring models using scikit-learn and MLflow.
- Delivered an observability-as-a-service model for rapid client onboarding and infrastructure integration.
- Reduced incident detection and root cause analysis time by 40%, improving operational uptime and SLA adherence.
- Presented architectural roadmaps, scaling benchmarks, and modernization strategies to executive stakeholders.
- Mentored engineers on streaming architecture, data reliability, and Kubernetes scaling best practices.
- Delivered a globally distributed analytics solution providing 24/7 monitoring across 10+ data centers.

Walmart - Chicago, Illinois

Data Engineer (Feb 2018 – May 2022)

Project: Enterprise Data Lake & ETL Modernization

- Led the transformation of Walmart's fragmented data environment into a unified AWS-based data lake serving analytics, ML, and supply chain optimization.
- Re-architected ingestion pipelines with Kafka, Spark, and Airflow, replacing legacy Teradata ETL flows that were slow and error-prone.
- Developed modular Python ingestion frameworks for ingesting POS, e-commerce, and logistics data across 50+ sources.
- Introduced event-driven CDC pipelines via Debezium + Kafka Connect, enabling near real-time data synchronization.
- Defined a multi-zone lake architecture (raw → refined → curated), enforcing clear boundaries for governance and lineage.
- Designed schema evolution workflows using Hive Metastore initially, later transitioning to AWS Glue Catalog for unified discovery.
- Optimized Spark jobs through adaptive partitioning, predicate pushdown, and file compaction, reducing job duration by half.
- Implemented dbt transformations (post-2020) to standardize SQL logic across analytics teams and eliminate redundant ETL scripts.
- Built Airflow DAG observability layers with Prometheus metrics and Grafana dashboards for run latency and SLA tracking.
- Introduced Great Expectations to automate dataset validation and eliminate silent data quality regressions.
- Collaborated with business analysts to unify inventory, pricing, and customer profiles, enabling 360° analytics views.
- Developed PySpark-based feature pipelines for demand forecasting models consumed by data science teams.
- Authored Terraform-based CI/CD blueprints that automated DAG deployments across multiple AWS environments.
- Designed data access and encryption policies meeting internal governance and compliance requirements.
- Built data lineage automation using Airflow metadata tables and Glue crawler outputs for end-to-end transparency.
- Consolidated siloed departmental marts into a single governed Redshift warehouse, cutting report refresh times by 60%.
- Worked with platform engineering to implement cost optimization strategies in S3 and Redshift storage layers.
- Led knowledge-sharing workshops on Airflow orchestration, PySpark optimization, and dbt modularity.
- Delivered a scalable, monitored, and self-healing data ecosystem supporting 20+ business domains and 2,000+ daily users.

First Citizens Bank – Mount Laurel, New Jersey

Data Analyst (Sep 2016 – Jan 2018)

- Owned design and build of a risk data pipeline unifying customer, credit, and deposit data from 12+ legacy systems.
- Implemented daily and hourly ETL pipelines in Python and SQL, automating what were previously manual report runs.
- Introduced the bank's first orchestration framework using early Apache Airflow for loan performance analytics.
- Built YAML-driven configuration layers for reusable ingestion logic — a pattern later adopted by multiple teams.
- Integrated credit bureau APIs and CRM data streams into a single scoring-ready warehouse on Redshift.
- Designed schema versioning using Protobuf, ensuring consistency and backward compatibility of risk models.
- Developed validation and exception logging frameworks using Python, ELK Stack, and structured audit logs.
- Implemented Kafka-based near real-time transaction streaming, feeding live updates to scoring and fraud systems.
- Used Spark and Pandas for credit segmentation, behavioral clustering, and exposure computation.
- Partnered with modelers to embed PD/LGD scoring into production workflows and automate score refreshes.
- Optimized PostgreSQL and Redshift queries handling 500M+ transaction records with indexed joins and partitioning.
- Deployed Jenkins pipelines for ETL deployments, Airflow DAG sync, and configuration promotion.
- Created Grafana dashboards (fed by ELK metrics) tracking pipeline freshness, latency, and anomaly counts.
- Developed checksum-based reconciliation to guarantee data completeness and regulatory traceability.
- Authored data access policies via AWS IAM roles, tightening control over sensitive financial data.
- Documented pipeline logic, schema diagrams, and lineage automatically to Confluence for audit readiness.
- Reduced credit scoring latency from 48 hours to under 2 hours, directly impacting underwriting speed.
- Delivered interactive Power BI dashboards enabling executives to monitor exposure, delinquency, and performance trends.
- Established the foundation of a data-driven risk analytics platform that evolved into a company-wide asset.

Microsoft – Chicago, Illinois

Big Data developer (Sep 2015 – Sep 2016)

- Assisted in developing large-scale data ingestion pipelines using Python, SQL, and the Hadoop ecosystem.
- Integrated structured and semi-structured data sources into Azure-based data lakes for analytics readiness.
- Developed and optimized ETL scripts for telemetry and operational data transformation.
- Worked with Hive and early Spark prototypes for batch aggregation and large-scale data analysis.
- Automated data validation and schema synchronization tasks with Python and SQL utilities.
- Implemented data profiling and cleansing scripts to improve accuracy across source systems.
- Contributed to partitioning and storage optimization for high-volume data tables.
- Created basic data lineage documentation and supported metadata tagging for analytics teams.
- Developed prototype dashboard visualizations for monitoring ingestion and job performance.
- Supported migration of legacy ETL jobs into Azure Data Factory and Power BI reporting environments.